

Лабораторная работа № 2

Измерение и тестирование пропускной способности сети. Интерактивный эксперимент

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Основной целью работы является знакомство с инструментом для измерения пропускной способности сети в режиме реального времени — iPerf3, а также получение навыков проведения интерактивного эксперимента по измерению пропускной способности моделируемой сети в среде Mininet.

```
mininet@mininet-vn:~$ cd /tmp
mininet@mininet-vn:/tmp$ git clone https://github.com/ekfoury/iperf3_plotter.git
Cloning into 'iperf3_plotter' ...
remote: Enumerating objects: 74, done.
remote: Total 74 (delta 0), reused 0 (delta 0), pack-reused 74 (from 1)
Unpacking objects: 100% (74/74), 100.09 KiB | 636.00 KiB/s, done.
mininet@mininet-vn:/tmp$ cd /tmp/iperf3_plotter
mininet@mininet-vn:/tmp/iperf3_plotter$ sudo cp plot_* /usr/bin
mininet@mininet-vn:/tmp/iperf3_plotter$ sudo cp *.sh /usr/bin
mininet@mininet-vn:/tmp/iperf3_plotter$
```

Рис. 1: Развертывание iperf3_plotter

Выполнение лабораторной работы

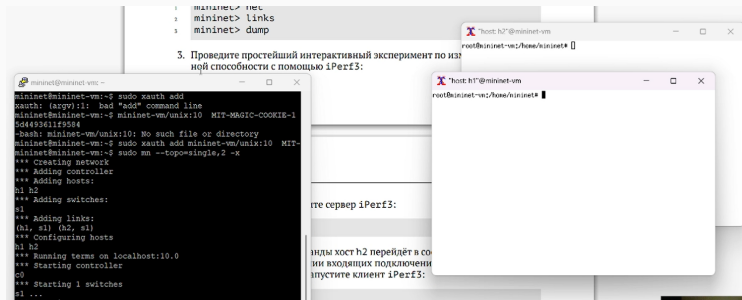


Рис. 2: Задание простейшей топологии

```
(h1, s1) (h2, s1)
*** Configuring hosts
h1 h2
*** Running terms on localhost:10.0
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Starting CLI:
mininet> net
h1 h1-eth0:s1-eth1
h2 h2-eth0:s1-eth2
s1 lo: s1-eth1:h1-eth0 s1-eth2:h2-eth0
c0
mininet> links
h1-eth0<->s1-eth1 (OK OK)
h2-eth0<->s1-eth2 (OK OK)
mininet> dump
<Host h1: h1-eth0:10.0.0.1 pid=6643>
<Host h2: h2-eth0:10.0.0.2 pid=6645>
<OVSSwitch s1: lo:127.0.0.1,s1-eth1:None,s1-eth2:None pid=6650>
<Controller c0: 127.0.0.1:6653 pid=6636>
mininet> 
```

Выполнение лабораторной работы

необходимо в терминале хоста h2 нажать `[Ctrl]+[C]`.

- Проанализируйте полученный в результате выполнения тест, отобразившийся как на клиенте, так и на сервере iPerf3 следующие данные:
 - ID: идентификационный номер соединения.
 - интервал (Interval): временной интервал для периодичности пропускной способности (по умолчанию временной интервал 1 сек).

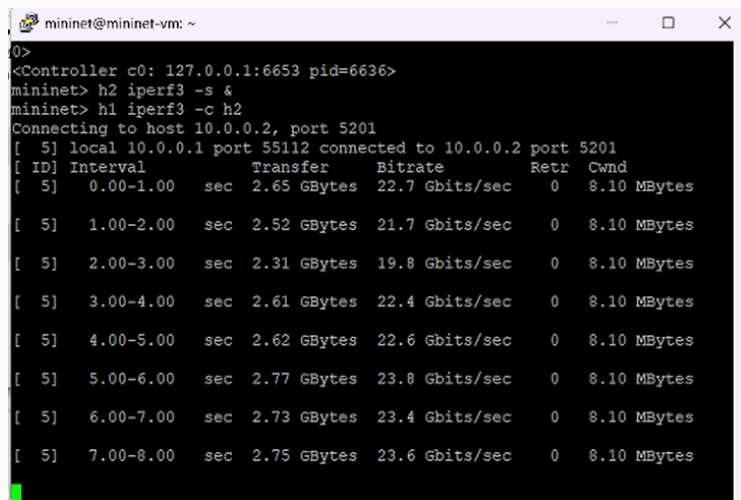
```
root@mininet-vx:/home/mininet# iperf3 -c 10.0.0.2
Connecting to host 10.0.0.2, port 5201
[ 7] local 10.0.0.1 port 55108 connected to 10.0.0.2 port 5201
[ ID] Interval      Transfer      Bitrate      Retr      Cwnd
[ 7] 0.00-1.00 sec  2.20 GBytes  18.8 Gbits/sec  0         8.17 MBytes
[ 7] 1.00-2.00 sec  2.53 GBytes  22.2 Gbits/sec  0         8.17 MBytes
[ 7] 2.00-3.00 sec  2.74 GBytes  23.6 Gbits/sec  0         8.17 MBytes
[ 7] 3.00-4.00 sec  2.33 GBytes  20.1 Gbits/sec  0         8.17 MBytes
[ 7] 4.00-5.00 sec  2.05 GBytes  17.6 Gbits/sec  0         8.17 MBytes
[ 7] 5.00-6.00 sec  2.52 GBytes  21.7 Gbits/sec  0         8.17 MBytes
[ 7] 6.00-7.00 sec  1.67 GBytes  14.4 Gbits/sec  0         8.17 MBytes
[ 7] 7.00-8.00 sec  1.48 GBytes  12.7 Gbits/sec  0         8.17 MBytes
[ 7] 8.00-9.00 sec  1.14 GBytes  9.82 Gbits/sec  0         8.17 MBytes
[ 7] 9.00-10.00 sec 1.56 GBytes  15.4 Gbits/sec  0         8.17 MBytes
-----
[ ID] Interval      Transfer      Bitrate      Retr      sender receiver
[ 7] 0.00-10.00 sec  20.3 GBytes  17.4 Gbits/sec  0
[ 7] 0.00-10.00 sec  20.3 GBytes  17.4 Gbits/sec  0

iperf Done.
root@mininet-vx:/home/mininet#
```

```
root@mininet-vx:/home/mininet# iperf3 -s
warning: this system does not seem to support IPv6 - trying IPv4
Server listening on 5201
Accepted connection from 10.0.0.1, port 55108
[ 7] local 10.0.0.2 port 5201 connected to 10.0.0.1 port 55108
[ ID] Interval      Transfer      Bitrate
[ 7] 0.00-1.00 sec  2.19 GBytes  18.8 Gbits/sec
[ 7] 1.00-2.00 sec  2.59 GBytes  22.2 Gbits/sec
[ 7] 2.00-3.00 sec  2.73 GBytes  23.5 Gbits/sec
[ 7] 3.00-4.00 sec  2.34 GBytes  20.1 Gbits/sec
[ 7] 4.00-5.00 sec  2.05 GBytes  17.6 Gbits/sec
[ 7] 5.00-6.00 sec  2.52 GBytes  21.7 Gbits/sec
[ 7] 6.00-7.00 sec  1.67 GBytes  14.4 Gbits/sec
[ 7] 7.00-8.00 sec  1.48 GBytes  12.7 Gbits/sec
[ 7] 8.00-9.00 sec  1.14 GBytes  9.82 Gbits/sec
[ 7] 9.00-10.00 sec 1.56 GBytes  15.4 Gbits/sec
[ 7] 10.00-10.00 sec 320 KBytes  575 Mbits/sec
-----
[ ID] Interval      Transfer      Bitrate
[ 7] 0.00-10.00 sec  20.3 GBytes  17.4 Gbits/sec

Server listening on 5201
```

Рис. 4: Тестовое соединение между хостами



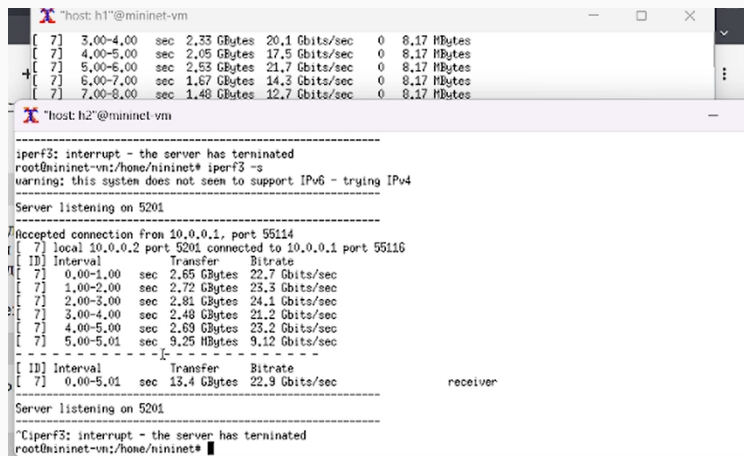
```
mininet@mininet-vm: ~
0>
<Controller c0: 127.0.0.1:6653 pid=6636>
mininet> h2 iperf3 -s &
mininet> h1 iperf3 -c h2
Connecting to host 10.0.0.2, port 5201
[ 5] local 10.0.0.1 port 55112 connected to 10.0.0.2 port 5201
[ ID] Interval          Transfer      Bitrate      Retr  Cwnd
[ 5]  0.00-1.00      sec  2.65 GBytes  22.7 Gbits/sec    0   8.10 MBytes
[ 5]  1.00-2.00      sec  2.52 GBytes  21.7 Gbits/sec    0   8.10 MBytes
[ 5]  2.00-3.00      sec  2.31 GBytes  19.8 Gbits/sec    0   8.10 MBytes
[ 5]  3.00-4.00      sec  2.61 GBytes  22.4 Gbits/sec    0   8.10 MBytes
[ 5]  4.00-5.00      sec  2.62 GBytes  22.6 Gbits/sec    0   8.10 MBytes
[ 5]  5.00-6.00      sec  2.77 GBytes  23.8 Gbits/sec    0   8.10 MBytes
[ 5]  6.00-7.00      sec  2.73 GBytes  23.4 Gbits/sec    0   8.10 MBytes
[ 5]  7.00-8.00      sec  2.75 GBytes  23.6 Gbits/sec    0   8.10 MBytes
```

Рис. 5: Эксперимент в интерфейсе mininet

```
mininet> h2 killall iperf3  
warning: this system does not seem to support IPv6 - t  
rying IPv4  
iperf3: error - unable to start listener for connectio  
ns: Address already in use  
iperf3: exiting  
mininet>
```

Рис. 6: Завершение процесса на сервере

Выполнение лабораторной работы



```
"host: h1"@mininet-vm
[ 7] 3.00-4.00 sec 2.33 GBytes 20.1 Gbits/sec 0 8.17 MBytes
[ 7] 4.00-5.00 sec 2.05 GBytes 17.5 Gbits/sec 0 8.17 MBytes
+ [ 7] 5.00-6.00 sec 2.53 GBytes 21.7 Gbits/sec 0 8.17 MBytes
[ 7] 6.00-7.00 sec 1.67 GBytes 14.3 Gbits/sec 0 8.17 MBytes
[ 7] 7.00-8.00 sec 1.48 GBytes 12.7 Gbits/sec 0 8.17 MBytes

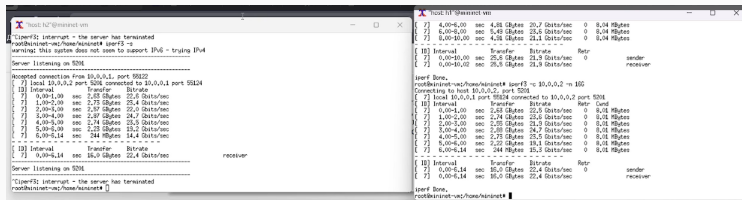
"host: h2"@mininet-vm
-----
iperf3: interrupt - the server has terminated
root@mininet-vm:/home/mininet# iperf3 -s
warning: this system does not seem to support IPv6 - trying IPv4
-----
Server listening on 5201
^
Accepted connection from 10.0.0.1, port 55114
[ 7] local 10.0.0.2 port 5201 connected to 10.0.0.1 port 55116
[ ID] Interval            Transfer        Bitrate
[ 7] 0.00-1.00 sec 2.65 GBytes 22.7 Gbits/sec
[ 7] 1.00-2.00 sec 2.72 GBytes 23.3 Gbits/sec
[ 7] 2.00-3.00 sec 2.81 GBytes 24.1 Gbits/sec
[ 7] 3.00-4.00 sec 2.48 GBytes 21.2 Gbits/sec
[ 7] 4.00-5.00 sec 2.69 GBytes 23.2 Gbits/sec
[ 7] 5.00-5.01 sec 9.25 MBytes 9.12 Gbits/sec
-----
[ ID] Interval            Transfer        Bitrate
[ 7] 0.00-5.01 sec 13.4 GBytes 22.9 Gbits/sec
-----
receiver
Server listening on 5201
^Ciperf3: interrupt - the server has terminated
root@mininet-vm:/home/mininet#
```

Рис. 7: Указание периода времени передачи

Выполнение лабораторной работы

```
"host: h2"@mininet-vm
Server listening on 5201
-----
^Ciperf3: interrupt - the server has terminated
root@mininet-vm:/home/mininet# iperf3 -s -i 2
warning: this system does not seem to support IPv6 - trying IPv4
Server listening on 5201
-----
Accepted connection from 10.0.0.1, port 55118
[ 7] local 10.0.0.2 port 5201 connected to 10.0.0.1 port 55120
[ ID] Interval      Transfer    Bitrate
[ 7] 0.00-2.00  sec  5.29 GBytes  22.7 Gbits/sec
[ 7] 2.00-4.00  sec  5.06 GBytes  21.7 Gbits/sec
[ 7] 4.00-6.00  sec  4.81 GBytes  20.6 Gbits/sec
[ 7] 6.00-8.00  sec  5.48 GBytes  23.5 Gbits/sec
[ 7] 8.00-10.00 sec  4.91 GBytes  21.1 Gbits/sec
[ 7] 10.00-10.02 sec  6.88 MBytes  3.05 Gbits/sec
-----
[ ID] Interval      Transfer    Bitrate
[ 7] 0.00-10.02  sec  25.5 GBytes  21.9 Gbits/sec
-----
Server listening on 5201
-----
```

Рис. 8: Выполнения теста пропускной способности с 2-секундным интервалом



```
root@mininet-ws:/home/mininet# iperf3 -s
warning: this system does not seem to support IPv6 - trying IPv4
Server listening on 5201
Accepted connection from 10.0.0.1, port 59122
[ 7 ] local 10.0.0.2 port 5201 connected to 10.0.0.1 port 59124
[ ID ] Interval      Transfer      Bitrate
[ 7 ] 0.00-1.00    sec  2.85 GBytes  22.6 Gbits/sec
[ 7 ] 1.00-2.00    sec  2.75 GBytes  22.4 Gbits/sec
[ 7 ] 2.00-3.00    sec  2.57 GBytes  22.0 Gbits/sec
[ 7 ] 3.00-4.00    sec  2.87 GBytes  24.7 Gbits/sec
[ 7 ] 4.00-5.00    sec  2.74 GBytes  23.5 Gbits/sec
[ 7 ] 5.00-6.00    sec  2.25 GBytes  19.3 Gbits/sec
[ 7 ] 6.00-6.14    sec   244 KBytes  14.4 Gbits/sec
[ ID ] Interval      Transfer      Bitrate
[ 7 ] 0.00-6.14    sec  16.0 GBytes  22.4 Gbits/sec
Server listening on 5201
*iperf3: Interrupt - the server has terminated
root@mininet-ws:/home/mininet#

root@mininet-ws:/home/mininet# iperf3 -c 10.0.0.2 -m 100
Connecting to host 10.0.0.2, port 5201
[ 7 ] local 10.0.0.1 port 59124 connected to 10.0.0.2 port 5201
[ ID ] Interval      Transfer      Bitrate      Retr      Cwnd
[ 7 ] 0.00-1.00    sec  2.63 GBytes  22.5 Gbits/sec  0      8.01 MBytes
[ 7 ] 1.00-2.00    sec  2.74 GBytes  23.6 Gbits/sec  0      8.01 MBytes
[ 7 ] 2.00-3.00    sec  2.25 GBytes  23.9 Gbits/sec  0      8.01 MBytes
[ 7 ] 3.00-4.00    sec  2.88 GBytes  24.7 Gbits/sec  0      8.01 MBytes
[ 7 ] 4.00-5.00    sec  2.73 GBytes  23.5 Gbits/sec  0      8.01 MBytes
[ 7 ] 5.00-6.00    sec  2.22 GBytes  19.1 Gbits/sec  0      8.01 MBytes
[ 7 ] 6.00-6.14    sec   244 KBytes  15.5 Gbits/sec  0      8.01 MBytes
[ ID ] Interval      Transfer      Bitrate      Retr      sender receiver
[ 7 ] 0.00-6.14    sec  16.0 GBytes  22.4 Gbits/sec  0
[ 7 ] 0.00-6.14    sec  16.0 GBytes  22.4 Gbits/sec  0      sender receiver
iperf3 Done.
root@mininet-ws:/home/mininet#
```

Рис. 9: Задание в тесте определённого объёма данных

Выполнение лабораторной работы

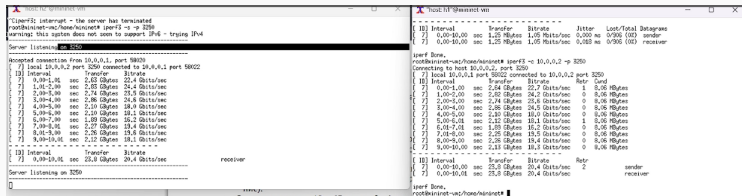
```
root@kali:~/iperfnet-src#
root@kali:~/iperfnet-src# iperf3 -s
warning: this system does not seem to support IPv6 - trying IPv4
Server listening on 5201

Accepted connection from 10.0.0.1, port 55126
[ I ] local 10.0.0.2 port 5201 connected to 10.0.0.1 port 44378
[ I ] Interval      Transfer      Bitrate      Jitter      Lost/total Datagrams
[ I ] 0.00-1.00 sec  129 KBytes  1.06 Mbits/sec  0.427 ms  0/90 (0%)
[ I ] 1.00-2.00 sec  127 KBytes  1.04 Mbits/sec  0.426 ms  0/90 (0%)
[ I ] 2.00-3.00 sec  129 KBytes  1.05 Mbits/sec  0.445 ms  0/91 (0%)
[ I ] 3.00-4.00 sec  127 KBytes  1.04 Mbits/sec  0.462 ms  0/90 (0%)
[ I ] 4.00-5.00 sec  129 KBytes  1.05 Mbits/sec  0.456 ms  0/91 (0%)
[ I ] 5.00-6.00 sec  127 KBytes  1.04 Mbits/sec  0.458 ms  0/90 (0%)
[ I ] 6.00-7.00 sec  129 KBytes  1.05 Mbits/sec  0.427 ms  0/91 (0%)
[ I ] 7.00-8.00 sec  127 KBytes  1.04 Mbits/sec  0.425 ms  0/90 (0%)
[ I ] 8.00-9.00 sec  127 KBytes  1.03 Mbits/sec  0.418 ms  0/90 (0%)
[ I ] 9.00-10.00 sec 129 KBytes  1.06 Mbits/sec  0.418 ms  0/91 (0%)
[ I ] 10.00-10.00 sec 1.41 KBytes  5.72 Mbits/sec  0.418 ms  0/1 (0%)
[ I ] Interval      Transfer      Bitrate      Jitter      Lost/total Datagrams
[ I ] 0.00-10.00 sec  1.29 MBytes  1.05 Mbits/sec  0.428 ms  0/906 (0%) receiver
Server listening on 5201

iperf Done.
root@kali:~/iperfnet-src# iperf3 -c 10.0.0.2 -u
Connecting to host 10.0.0.1, port 5201
[ I ] local 10.0.0.1 port 4430 connected to 10.0.0.2 port 5201
[ I ] Interval      Transfer      Bitrate      Jitter      Total Datagrams
[ I ] 0.00-1.00 sec  129 KBytes  1.05 Mbits/sec  31
[ I ] 1.00-2.00 sec  127 KBytes  1.04 Mbits/sec  30
[ I ] 2.00-3.00 sec  129 KBytes  1.05 Mbits/sec  31
[ I ] 3.00-4.00 sec  129 KBytes  1.05 Mbits/sec  31
[ I ] 4.00-5.00 sec  127 KBytes  1.04 Mbits/sec  30
[ I ] 5.00-6.00 sec  127 KBytes  1.04 Mbits/sec  30
[ I ] 6.00-7.00 sec  129 KBytes  1.05 Mbits/sec  31
[ I ] 7.00-8.00 sec  129 KBytes  1.05 Mbits/sec  31
[ I ] 8.00-9.00 sec  128 KBytes  1.02 Mbits/sec  30
[ I ] 9.00-10.00 sec  130 KBytes  1.07 Mbits/sec  32
[ I ] Interval      Transfer      Bitrate      Jitter      Lost/Total Datagrams
[ I ] 0.00-10.00 sec  1.25 MBytes  1.05 Mbits/sec  0.000 ms  0/906 (0%) sender
[ I ] 0.00-10.00 sec  1.25 MBytes  1.05 Mbits/sec  0.088 ms  0/906 (0%) receiver
iperf Done.
root@kali:~/iperfnet-src#
```

Рис. 10: Изменение протокола передачи данных

Выполнение лабораторной работы



```
root@minet-uc:/home/minet# iperf3 -s -p 3250
warning: this system does not seem to support IPv6 - trying IPv4
Server listening on 3250

Accepted connection from 10.0.0.1, port 58020
[ ID] Interval      Transfer      Bitrate
[ 7] 0.00-1.00 sec  2.63 GBytes  22.4 Gbits/sec
[ 7] 1.00-2.00 sec  2.65 GBytes  24.4 Gbits/sec
[ 7] 2.00-3.00 sec  2.74 GBytes  23.5 Gbits/sec
[ 7] 3.00-4.00 sec  2.86 GBytes  24.6 Gbits/sec
[ 7] 4.00-5.00 sec  2.10 GBytes  18.0 Gbits/sec
[ 7] 5.00-6.00 sec  2.10 GBytes  18.1 Gbits/sec
[ 7] 6.00-7.00 sec  1.89 GBytes  16.2 Gbits/sec
[ 7] 7.00-8.00 sec  2.27 GBytes  19.4 Gbits/sec
[ 7] 8.00-9.00 sec  2.25 GBytes  19.6 Gbits/sec
[ 7] 9.00-10.00 sec 2.12 GBytes  18.1 Gbits/sec
[ ID] Interval      Transfer      Bitrate
[ 7] 0.00-10.00 sec 23.0 GBytes  20.4 Gbits/sec
Server listening on 3250

iperf Done.
root@minet-uc:/home/minet# iperf3 -c 10.0.0.2 -p 3250
Connecting to host 10.0.0.1 port 3250
[ ID] Interval      Transfer      Bitrate  Retr  Cwnd
[ 7] 0.00-1.00 sec  2.64 GBytes  22.7 Gbits/sec  1  8.06 MBytes
[ 7] 1.00-2.00 sec  2.82 GBytes  24.2 Gbits/sec  0  8.06 MBytes
[ 7] 2.00-3.00 sec  2.74 GBytes  23.6 Gbits/sec  0  8.06 MBytes
[ 7] 3.00-4.00 sec  2.86 GBytes  24.6 Gbits/sec  0  8.06 MBytes
[ 7] 4.00-5.00 sec  2.80 GBytes  24.0 Gbits/sec  0  8.06 MBytes
[ 7] 5.00-6.00 sec  2.12 GBytes  18.1 Gbits/sec  1  8.06 MBytes
[ 7] 6.00-7.00 sec  1.89 GBytes  16.2 Gbits/sec  0  8.06 MBytes
[ 7] 7.00-8.00 sec  2.25 GBytes  19.5 Gbits/sec  0  8.06 MBytes
[ 7] 8.00-9.00 sec  2.25 GBytes  19.4 Gbits/sec  0  8.06 MBytes
[ 7] 9.00-10.00 sec 2.11 GBytes  18.1 Gbits/sec  0  8.06 MBytes
[ ID] Interval      Transfer      Bitrate  Retr
[ 7] 0.00-10.00 sec 23.0 GBytes  20.4 Gbits/sec  2
[ 7] 0.00-10.00 sec 23.0 GBytes  20.4 Gbits/sec
iperf Done.
root@minet-uc:/home/minet#
```

Рис. 11: Изменение номера порта для отправки/получения пакетов

```
Server listening on 8269
.....
^Ciperrf3: interrupt - the server has terminated
root@binnet-wz/home/kinninet# iperrf3 -s 1
warning: this system does not seem to support IPv6 - trying IPv4

Server listening on 8269
.....
Accepted connection from 10.0.0.1, port 55132
[+] local 10.0.0.2 port 5201 connected to 10.0.0.1 port 55134
[+] Interval Transfer Bitrate
[+] 0.0-1.0 sec 2.76 Gbytes 25.7 Gbits/sec
[+] 1.0-2.0 sec 2.80 Gbytes 24.1 Gbits/sec
[+] 2.0-3.0 sec 2.82 Gbytes 24.3 Gbits/sec
[+] 3.0-4.0 sec 2.72 Gbytes 24.3 Gbits/sec
[+] 4.0-5.0 sec 2.81 Gbytes 24.1 Gbits/sec
[+] 5.0-6.0 sec 2.75 Gbytes 23.6 Gbits/sec
[+] 6.0-7.0 sec 2.79 Gbytes 24.0 Gbits/sec
[+] 7.0-8.0 sec 2.57 Gbytes 22.1 Gbits/sec
[+] 8.0-9.0 sec 2.90 Gbytes 24.1 Gbits/sec
[+] 9.0-10.0 sec 2.82 Gbytes 24.2 Gbits/sec
[+] 10.0-10.0 sec 20.3 Gbytes 11.4 Gbits/sec

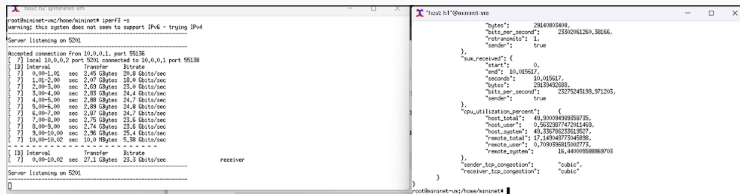
[+] Interval Transfer Bitrate
[+] 0.0-10.0 sec 27.6 Gbytes 23.7 Gbits/sec
root@binnet-wz/home/kinninet#

iperrf3:
root@binnet-wz/home/kinninet# iperrf3 -s 10.0.0.2
connecting to host 10.0.0.2, port 5201
[+] local 10.0.0.1 port 55134 connected to 10.0.0.2 port 5201
[+] Interval Transfer Bitrate Retr Cond
[+] 0.0-1.0 sec 2.76 Gbytes 25.6 Gbits/sec 0 8.37 Mbytes
[+] 1.0-2.0 sec 2.81 Gbytes 24.2 Gbits/sec 0 8.37 Mbytes
[+] 2.0-3.0 sec 2.81 Gbytes 24.1 Gbits/sec 0 8.37 Mbytes
[+] 3.0-4.0 sec 2.72 Gbytes 23.3 Gbits/sec 0 8.37 Mbytes
[+] 4.0-5.0 sec 2.81 Gbytes 24.1 Gbits/sec 0 8.37 Mbytes
[+] 5.0-6.0 sec 2.78 Gbytes 23.6 Gbits/sec 0 8.37 Mbytes
[+] 6.0-7.0 sec 2.78 Gbytes 23.8 Gbits/sec 0 8.37 Mbytes
[+] 7.0-8.0 sec 2.55 Gbytes 22.3 Gbits/sec 0 8.37 Mbytes
[+] 8.0-9.0 sec 2.78 Gbytes 23.9 Gbits/sec 0 8.37 Mbytes
[+] 9.0-10.0 sec 2.85 Gbytes 24.2 Gbits/sec 0 8.37 Mbytes

[+] Interval Transfer Bitrate Retr
[+] 0.0-10.0 sec 27.6 Gbytes 23.7 Gbits/sec 0
[+] 0.0-10.0 sec 27.6 Gbytes 23.7 Gbits/sec 0
root@binnet-wz/home/kinninet#
```

Рис. 12: Параметр обработки данных только от одного клиента с остановкой сервера по завершении теста

Выполнение лабораторной работы



The image shows two terminal windows. The left window, titled 'root@kali:~#', shows the execution of 'iperf3 -s' on a Kali Linux machine. It displays a warning about IPv6 support and then shows a list of accepted connections from 10.0.0.1. The right window, titled 'root@kali:~#', shows the output of 'cat /dev/null' which is a JSON file containing the test results. The JSON file includes fields for 'bytes', 'bits_per_second', 'retransmits', 'sender', 'sum_received', 'start', 'end', 'seconds', 'bytes', 'bits_per_second', 'cpu_utilization_percent', 'host_total', 'host_user', 'host_system', 'remote_total', 'remote_user', 'remote_system', 'sender_tcp_congestion', and 'receiver_tcp_congestion'.

```
root@kali:~# iperf3 -s
warning: this system does not seem to support IPv6 - trying IPv4
Server listening on 5201

Accepted connection from 10.0.0.1, port 55136
[1] local 10.0.0.2 port 5201 connected to 10.0.0.1 port 55136
[1] Interval Transfer Bitrate
[1] 0.00-1.00 sec 2.45 GBytes 20.8 Gbits/sec
[1] 1.00-2.00 sec 2.07 GBytes 18.0 Gbits/sec
[1] 2.00-3.00 sec 2.03 GBytes 17.6 Gbits/sec
[1] 3.00-4.00 sec 2.03 GBytes 17.4 Gbits/sec
[1] 4.00-5.00 sec 2.89 GBytes 24.7 Gbits/sec
[1] 5.00-6.00 sec 2.89 GBytes 24.6 Gbits/sec
[1] 6.00-7.00 sec 2.07 GBytes 18.0 Gbits/sec
[1] 7.00-8.00 sec 2.75 GBytes 23.4 Gbits/sec
[1] 8.00-9.00 sec 2.74 GBytes 23.4 Gbits/sec
[1] 9.00-10.00 sec 2.96 GBytes 25.4 Gbits/sec
[1] 10.00-10.00 sec 10.0 MBytes 5.38 Gbits/sec
[1] Interval Transfer Bitrate
[1] 0.00-10.00 sec 27.1 GBytes 23.3 Gbits/sec
Server listening on 5201
0
```

```
root@kali:~# cat /dev/null
{
  "bytes": 2914083848,
  "bits_per_second": 2392961269.38666,
  "retransmits": 1,
  "sender": true
},
{
  "sum_received": {
    "start": 0,
    "end": 10.005617,
    "seconds": 10.015617,
    "bytes": 29129442393,
    "bits_per_second": 23275245139.371295,
    "sender": true
  },
  "cpu_utilization_percent": {
    "host_total": 49.300094939359735,
    "host_user": 0.563239774720114039,
    "host_system": 49.306796233119627,
    "remote_total": 17.1494877346898,
    "remote_user": 0.793939815002773,
    "remote_system": 16.4490958868705
  },
  "sender_tcp_congestion": "cubic",
  "receiver_tcp_congestion": "cubic"
}
```

Рис. 13: Экспорт результатов теста измерения пропускной способности iPerf3 в файл JSON

```
root@mininet-vx:/home/mininet# iperf3 -c 10.0.0.2 -J > /home/mininet/work/lab_iperf3/iperf_results.json
root@mininet-vx:/home/mininet# cd /home/mininet/work/lab_iperf3
root@mininet-vx:/home/mininet/work/lab_iperf3# ls -l
total 0
root@mininet-vx:/home/mininet/work/lab_iperf3#
```

Рис. 14: Проверка создания файла iperf_results.json

```
mininet@mininet-vm:~$ xauth list
mininet-vm/unix:11 MIT-MAGIC-COOKIE-1 f12cf0cb97ffa8bc5ec2f6f886a910fb
mininet-vm/unix:10 MIT-MAGIC-COOKIE-1 a701335a94d0cfa3245d4493611f9584
mininet@mininet-vm:~$ sudo xauth add
xauth: (argv):1: bad "add" command line
mininet@mininet-vm:~$ mininet-vm/unix:10 MIT-MAGIC-COOKIE-1 a701335a94d0cfa324
5d4493611f9584
```

Рис. 15: Исправление прав запуска X-соединения

```
mininet@mininet-vm:~/work/lab_iperf3$ ls -l
total 8
-rw-r--r-- 1 root root 7819 Sep 21 11:00 iperf_results.json
mininet@mininet-vm:~/work/lab_iperf3$ sudo chown -R mininet:mininet ~/work
mininet@mininet-vm:~/work/lab_iperf3$ ls -l
total 8
-rw-r--r-- 1 mininet mininet 7819 Sep 21 11:00 iperf_results.json
mininet@mininet-vm:~/work/lab_iperf3$ plot_iperf.sh iperf3_results.json
Error: iperf3_results.json is not a file. Quitting...
mininet@mininet-vm:~/work/lab_iperf3$ ls -l
total 8
-rw-r--r-- 1 mininet mininet 7819 Sep 21 11:00 iperf_results.json
mininet@mininet-vm:~/work/lab_iperf3$ plot_iperf.sh iperf_results.json
mininet@mininet-vm:~/work/lab_iperf3$ cd ~/work/lab_iperf3
mininet@mininet-vm:~/work/lab_iperf3$ ls -l
total 16
-rw-rw-r-- 1 mininet mininet 966 Sep 21 11:04 iperf.csv
-rw-r--r-- 1 mininet mininet 7819 Sep 21 11:00 iperf_results.json
drwxrwxr-x 2 mininet mininet 4096 Sep 21 11:04 results
mininet@mininet-vm:~/work/lab_iperf3$
```

Рис. 16: Визуализация результатов эксперимента

В результате выполнения данной лабораторной работы я познакомился с инструментом для измерения пропускной способности сети в режиме реального времени — iPerf3, а также получила навыки проведения интерактивного эксперимента по измерению пропускной способности моделируемой сети в среде Mininet.